

12th CIRP Conference on Photonic Technologies [LANE 2022], 4-8 September 2022, Fürth, Germany

Characterization of keyhole dynamics in laser welding of copper by means of high-speed synchrotron X-ray imaging

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Abstract

Laser welding of copper is of great importance for industrial applications, e.g., for manufacturing of electrical components such as hairpins. Solid state lasers are widely used due to the high power and beam brilliance, but the implementation can be challenging in terms of process instabilities and resulting weld defects, i.e., spatter and pore formation. In order to understand the formation thus avoid these defects, the development of the understanding of the keyhole and its interaction with the surrounding melt pool is required. In this paper, high-speed synchrotron X-ray imaging (frame rate: 20,000 Hz) demonstrated its capability to study the keyhole geometry to quantify and describe the dynamic behavior inside the workpiece. Thus, novel insights into fundamental processes are provided and a new methodic approach was introduced to describe the time-dependent behavior of the keyhole and its interaction with the melt pool during laser beam deep welding of copper (Cu-ETP/CW004A).

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Peer-review under responsibility of the international review committee of the 12th CIRP Conference on Photonic Technologies [LANE 2022]

Keywords: laser welding; copper; high-speed X-ray imaging; synchrotron

1. Introduction and state of the art

Laser welding is a key technology to achieve high quality electrical and mechanical connections. Due to the high energy inputs and precision, solid state lasers are generally used for laser welding to provide high efficiency of the process for most engineering materials. Typical solid-state lasers for welding are disk lasers and fiber lasers with wavelengths between 1,030 and 1,070 nm. However, the absorptivity of copper in this wavelength range is limited to 6–7 % [1]. In combination with the high thermal conductivity, laser welding of copper remains challenging. Copper can only be processed using deep penetration welding mode due to the aforementioned factors,

whereas the high intensity of the laser beam melts and vaporizes the material. The evaporation leads to deformation of the resulting melt pool and multiple reflections of the laser beam. A keyhole is formed due to this process [2]. The keyhole formation depends on a pressure balance between opening and closing pressures, consisting of recoil pressure p_{abl} , differential vapor pressure $\Delta(p_v - p_0)$, atmospheric pressure p_0 , surface tension pressure p_σ , hydrostatic pressure p_h and the hydrodynamic pressure p_{dyn} . (Equation 1) [3].

$$\underbrace{p_{abl} + \Delta(p_v - p_0)}_{\text{opening pressures}} = \underbrace{p_\sigma + p_h + p_{dyn}}_{\text{closing pressures}} \quad (1)$$

The pressure balance is affected by the energy input, the intensity distribution, the metal vapor, the melt flow and the atmospheric pressure among others [4]. The complexity of the pressure balance introduces fluctuations in the keyhole. These fluctuations occur periodically and in different spatial directions [5]. A model-based study showed that the fluctuations were stable at infinitesimally small amplitudes and it is calculated that the typical eigenfrequencies are up to 2 kHz [5]. However, experiments showed that several frequencies can be superpositioned, e.g. up to 13.5 kHz based on acoustic emissions [6], up to 10 kHz for spectral emissions [7] and up to 2 kHz for high-speed recordings accompanied by significant amplitudes of more than 100 μm of the keyhole aperture [8]. The higher frequencies and high amplitudes in experiments compared to model-based assumptions can be explained by instabilities of the keyhole [5] which causes the formation of pores and spatters in laser beam welding of copper [9]. The effect of the keyhole behavior on spatter and pore formation was investigated by X-ray imaging at 1 kHz [10]. However, due to the limitation in spatial and temporal resolution a deeper understanding of the keyhole behavior with regard to its fluctuations at high frequencies up to several kilohertz has not been studied.

The use of high-energy, brilliant synchrotron radiation can greatly improve the temporal and spatial resolution of the X-ray imaging which provide us with an opportunity to reveal novel insights in the high frequency behavior of the keyhole. The capabilities have been proven for additive manufacturing with acquisition rates up to 40 kHz [11].

This paper, for the first time, demonstrated the capabilities of in situ high-speed synchrotron imaging regarding a highly detailed visualization of the keyhole and its interactions with the surrounding melt pool during laser beam welding of copper. A novel methodological approach is introduced including new ways to describe the time-dependent behavior of the keyhole by means of geometrical parameters and to determine the oscillation frequencies in different spatial directions. The findings lead to new scientific questions in keyhole behavior control and offer starting points for a further improvement of the welding process.

2. Materials and methods

2.1. Laser welding setup

A fiber laser COHERENT HighLight FL8000-ARM and a stationary processing head Precitec YW52 were used for carrying out the welding process.

Table 1. Parameters of laser welding setup

Parameter	Symbol	Quantity and unit
Beam power	P_L	3,500 W
Wavelength	λ	1,070 nm
Fiber diameter of center beam	-	50 μm
Focal position	z_0	Copper surface
Rayleigh length	z_R	0.84 mm
Focal diameter	d_f	89 μm
Welding speed	v_w	10 m/min

The laser beam source was equipped with an adjustable ring mode (ARM) fiber, however, just the center beam fiber was used during the experiments. The specifications of the laser system are given in Table 1. The laser beam caustic was measured by a Primes FocusMonitor at a constant beam power of 300 W to avoid damaging the measurement system. The focal diameter d_f and the Rayleigh length z_R were determined based on the D86 beam width measurement method.

2.2. High-speed synchrotron imaging

High-speed synchrotron X-ray imaging was performed at the European Synchrotron Radiation Facility (ESRF) at Beamline ID19. The X-ray beam size was set to $4 \times 4 \text{ mm}^2$ with an energy $\leq 60 \text{ keV}$. Ce-doped $\text{Lu}_3\text{Al}_5\text{O}_{12}$ with a thickness of 250 μm from manufacturer Crytur was used as scintillator material, lens-coupled to a Photron SA-Z high-speed camera (5x/0.14 NA, propagation distance 4.5 m, framerate 20,000 Hz). A spatial resolution of 4 μm per pixel was reached within the high-speed recordings. A schematic of the setup is given in Fig. 1.

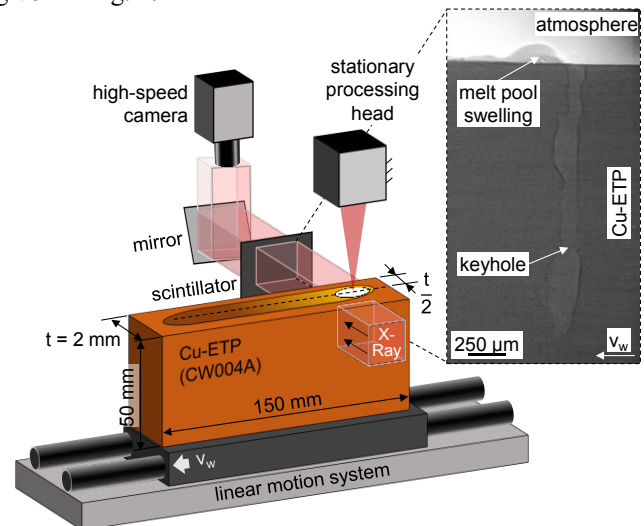


Fig. 1. Schematic of setup for high-speed synchrotron X-ray imaging.

Partial penetration welds were performed on copper specimens (Cu-ETP/CW004A) with a thickness t of 2 mm and a height of 50 mm. The beam was aligned to the center of the specimen at $t/2$. The welds were carried out as partial penetration welds at a constant welding speed of 10 m/min. The sample was moving while the processing head was stationary in order to ensure the recording of the keyhole during each whole experiment. An exemplary image is given in Fig. 2 that shows a section of the recorded area including the keyhole. The region above the copper surface was recorded to gain information on melt pool swelling and spatter formation. An increased grey value allows the detection of the keyhole due to a reduced attenuation of the beam travelling through the metal vapor compared to solid or liquid copper. The software Photron Fastcam Viewer 4 was used for image post-processing. The High Dynamic Range option of the software was applied to the images and minor corrections of brightness (+0.14) and contrast (+0.14) were performed to increase the grey value differences between the keyhole and its surroundings.

2.3. Data evaluation

A number of 256 consecutive frames were taken from the recordings and evaluated to demonstrate the capability of the data evaluation. This corresponds to a time interval of 12.8 ms, i.e. a frame was recorded every 0.05 ms. The time interval was selected so that it is in the first third of the weld after an almost constant weld penetration depth was reached. The start time of the interval considered is set to $\tau = 0$. The occurrence of massive weld defects, e.g. the complete ejection of the melt pool, does not occur in the time interval considered.

Different parameters were evaluated for each frame, i.e. time-dependent information were obtained for keyhole aperture at the surface d , keyhole depth t_d , the maximum of the keyhole bulging a and the depth of the maximum bulging t_a (see Fig. 2). The maximum bulging a represents the maximum width of the keyhole, i.e. deformations on the keyhole front are also considered. The formation of the ratio a/d provides information on the bulging related to the keyhole aperture and t_a/t_d on the location of the maximum bulging related to keyhole depth. This allows a time-dependent description to shape deviations of the keyhole contour to quantify the X-ray images.

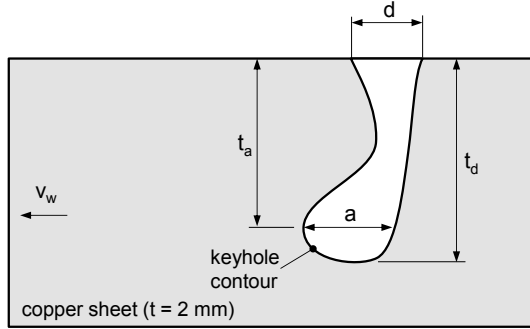


Fig. 2. Schematic of time-dependently evaluated keyhole parameters

Considering the consecutive frames and obtaining the mentioned parameters for each frame allowed a further data analysis using Fast Fourier Transform (FFT) to determine the oscillations of keyhole aperture d and keyhole depth t_d . Different frequencies were determined as peaks in the FFT. To verify the dominant frequency – since the sample size was limited to 256 frames – a sine fit was applied to the data. Equation 2 shows the sine fit $s(t)$ considering time τ , frequency f , mean absolute deviation MAD of the data as amplitude, median value $\tilde{d}$ as midline and phase shift φ_{opt} .

$$s(\tau) = MAD \cdot \sin(2\pi f\tau + \varphi_{opt}) + \tilde{d} \quad (2)$$

The median value $\tilde{d}$ and the mean absolute deviation MAD , respectively, were chosen to avoid overweighting outliers of the keyhole geometry for the sine fit. The phase shift φ_{opt} is not provided by the FFT. Therefore, a generalized reduced gradient (GRG) nonlinear solver with multiple starts was used to evaluate the lowest mean square error (MSE) while varying φ_{opt} for every peak frequency determined by the FFT. A comparison between these results allowed to identify the dominant frequency for keyhole aperture and depth based on the smallest reached MSE.

3. Results and discussion

3.1. Description of keyhole behavior

A brief overview of the keyhole behavior is given based on a time interval that reflects various effects typical for copper welding at these focal diameters and intensities.

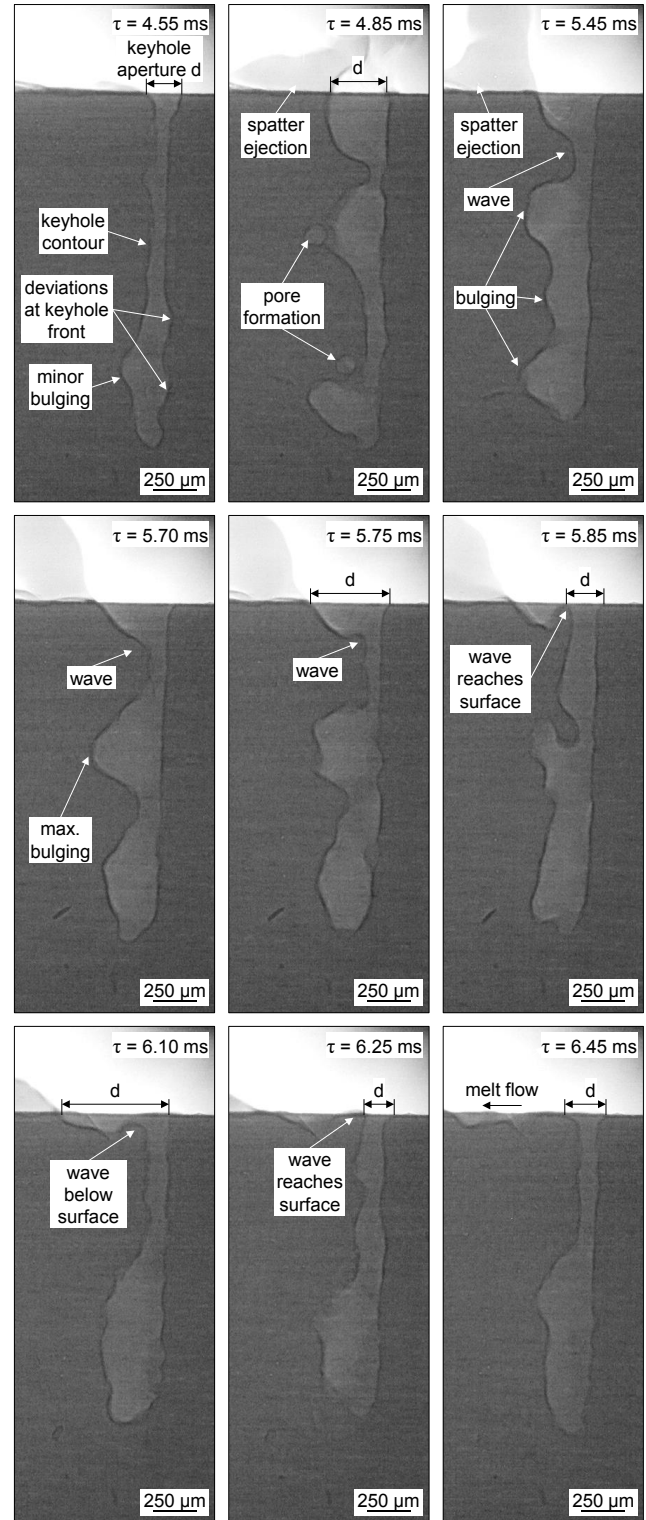


Fig. 3. Highspeed synchrotron X-ray images of the keyhole in copper

Fig. 3 shows the keyhole during laser welding of copper for different times and demonstrates the high spatial and temporal resolution of the synchrotron high-speed imaging. It should be noted that $\tau = 0$ ms corresponds to the start of the time interval considered and not to the start of the welding process (see also section 2.3). The image series shows a section from the middle range of the interval considered, representing different effects starting at $\tau = 4.55$ ms. The keyhole appeared relatively slim, as expected for the chosen focal diameter of approx. $89 \mu\text{m}$ and had a narrow aperture d at the surface. Irregularities of the keyhole geometry increased with the depth of the keyhole. It is noticeable that shape deviations occurred not only as minor bulging on the rear wall, but also on the keyhole front. In contrast to existing X-ray investigations of copper welds, e.g. in [9] or [10], the image quality and the temporal resolution was significantly improved.

The processes in the keyhole and the interaction between keyhole and surrounding melt pool can be resolved highly dynamic. This became obvious during the formation of spatters when the keyhole aperture was significantly enlarged due to the ejection of material ($\tau = 4.85$ ms). The keyhole geometry also changed in deeper regions at the same time, i.e. bulging increased at different positions and pores were pinched off from the keyhole at the bulges. The pores often went directly back into the keyhole due to the dynamic movement of the keyhole and were thus dissolved, i.e. they could no longer be found in the rear melt pool ($\tau = 5.45$ ms). The changes in keyhole geometry could be well traced by newly emerging bulges too, while the keyhole aperture remains widely open. Spatters created a wave that moved upward and reached the surface at $\tau = 5.85$ ms. Due to the melt pool dynamics, the crest of the wave sank down in the following before rising again ($\tau = 6.10$ – 6.25 ms). The wave then continued to flow in the direction of the backward melt pool traceable by the movement of the trough. During this time, the keyhole aperture d at the top was significantly increased and fluctuated strongly. It was defined for the evaluation that the area without melted material at the surface is considered as keyhole aperture d to fully capture the dynamic, even if the evaporation area is difficult to delineate from melt-flow effected areas.

3.2. Time-dependent quantification of imaging results

Different parameters were considered for quantifying the X-ray images. Fig. 4a shows the measured values of keyhole aperture d over time for 256 consecutive frames which is equivalent to a time interval of 12.8 ms. The values of d vary between $96 \mu\text{m}$ as minimum and $676 \mu\text{m}$ as maximum. The keyhole aperture fluctuated significantly at different times, e.g. a significant widening started from 4.5 ms, 9.1 ms or 10.3 ms onwards. The effect was strongly influenced by the melt pool flow as shown in Fig. 4.

An observation of the measured values of d in the frequency domain can be obtained by means of FFT (Fig. 4b). It was found that not a single frequency was formed, but multiple frequencies occurred, indicating superimposed vibrations of the keyhole. The peaks of the predominantly occurring frequencies were between 234 Hz and 1,563 Hz.

Based on the optimization algorithm and the determined phase shift, all peak frequencies were considered by applying the sine fit and its evaluation with respect to MSE as described in section 2.3. The table in Fig. 4b shows the frequencies and corresponding MSEs. The smallest MSE was obtained for 859 Hz and this frequency was transferred as sine fit in Fig. 4a including the optimized phase shift. A high agreement of this dominant frequency regarding the aperture fluctuation can be seen in the time domain, although it is a strong simplification due to the superimposed frequencies. For the observation of other time intervals, a different result might be obtained, also a small difference in the MSE to the frequency of 234 Hz or 625 Hz was noticeable.

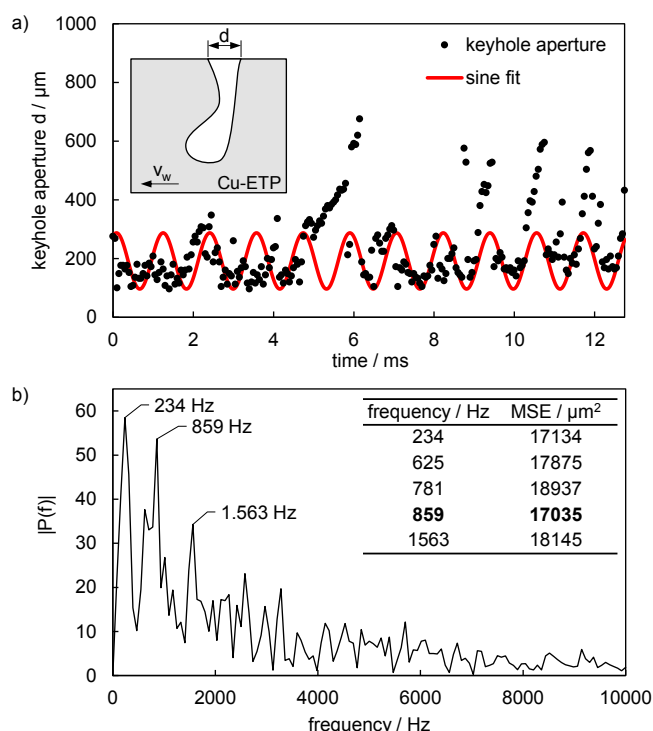


Fig. 4. a) Time-dependent behavior of keyhole aperture d and b) determined frequencies by means of FFT.

A similar procedure was chosen for the keyhole depth t_d . The value of t_d ranged between $1,433 \mu\text{m}$ as minimum and $2,088 \mu\text{m}$ as maximum value (see Fig. 5a). The keyhole depth changed rapidly by several tenths of a millimeter even for the short time intervals between several frames. It should be noted that the actual welding depth may vary less, since it depends on the thermal conduction between the melt pool adjacent to the keyhole and the solid material, which is more inert compared to the processes in the keyhole.

A consideration of the frequency domain revealed several peaks for the keyhole depth t_d as well, although these tended to occur at lower frequencies than for the keyhole aperture d . The lowest MSE regarding the keyhole depth t_d was obtained for a frequency of 234 Hz.

This result was transferred to Fig. 5a and plotted as a sine fit by considering the optimized phase shift (see section 2.3). Again, a good agreement between determined frequency and time domain was observed. Considering frequencies of keyhole aperture d and depth t_d in comparison based on the frequency

domains (Fig. 4b, Fig. 5b) revealed lower frequencies and fewer peaks for the keyhole depth. It was noticeable that the frequency of 234 Hz appears for both measured quantities considered. A mutual influence of radial and axial frequencies is conceivable and will be subject of further studies.

A complete collapse of the keyhole did not occur in the time range considered and the bottom of the keyhole was less affected by melt pool flows than the keyhole aperture. However, strong fluctuations occurred also at the keyhole bottom due to the time-varying energy input, although these were less evident in the keyhole depth and more in the bulging.

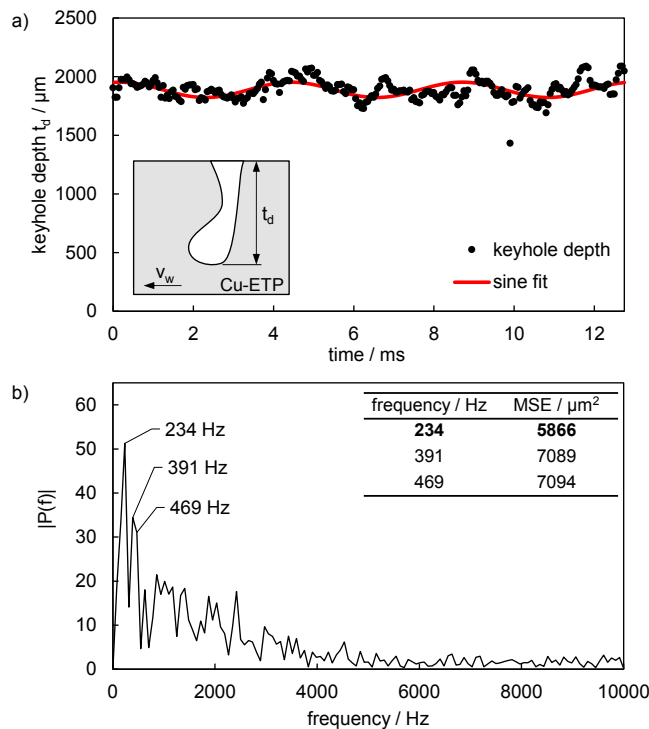


Fig. 5. a) Time-dependent behavior of keyhole depth t_d and b) determined frequencies by means of FFT.

3.3. Evaluation of keyhole bulging

The discussed frequencies of aperture d and depth t_d reflected the keyhole dynamics to some extent. Bulging had to be included as a well-known phenomenon for characterizing the keyhole behavior to obtain a holistic picture for laser welding of copper.

Fig. 6a shows the depth of the maximum bulging t_a related to keyhole depth t_d by forming the ratio t_a/t_d . For a value of 1, the largest bulge would be at the maximum keyhole depth. For times up to 4.4 ms, bulging occurs mainly in the bottom region of the keyhole, i.e. the ratio t_a/t_d reached values up to 0.94. The maximum bulging occurred at other locations as the fluctuations increased (noticeable also in Fig. 3 and Fig. 4a).

Very fast changes between the middle, the upper as well as the lower region of the keyhole occurred with respect to the maximum bulging. The fluctuation was also apparent for the ratio between maximum bulging a and keyhole aperture d (Fig. 6b). The bulging reached a ratio of up to 5.2 and varied strongly over time. The keyhole in the inside is on average twice as long compared to the aperture.

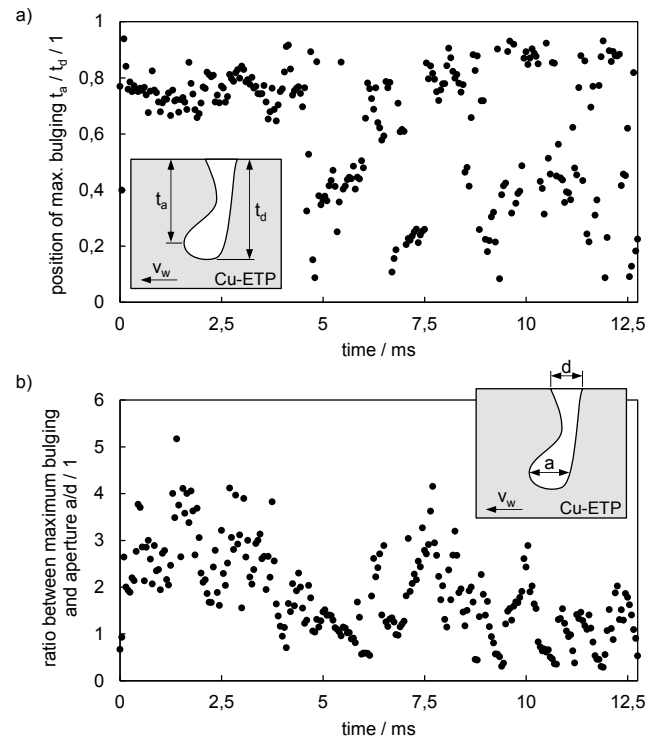


Fig. 6. a) Position of maximum bulging based on ratio t_a/t_d and b) ratio between maximum bulging and keyhole aperture a/d .

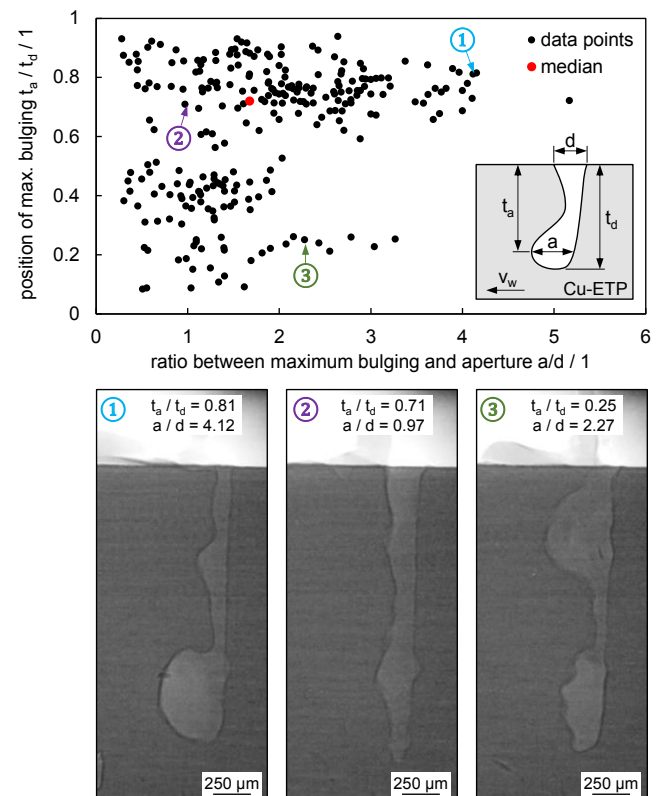


Fig. 7. Position of maximum bulging and related enlargement of the keyhole.

A comparative representation of both ratios is given in Fig. 7. The plot provides a comparison of the maximum enlargement of the bulge and its position in depth within the keyhole. It was noticeable that the bulging at the keyhole

bottom was in most cases more pronounced than in the middle or upper region. This can be explained by the reflection of the laser at the keyhole front, which preferably is reflected into the bottom region. A raytracing within the keyhole for different time steps based on the data obtained will provide further information on this. It should be noted, that in addition to the maximum bulging, further bulges could occur that were merely more limited in size (see Fig.4). The positions ①, ② and ③ represent this behavior for a strong bulging at the bottom, an almost straight keyhole, and a maximum bulge close to the surface. It should be considered that several bulges occur at different positions mostly and vary quickly over time.

4. Summary and outlook

High-speed synchrotron X-ray imaging demonstrated its capabilities for studying the keyhole behavior during laser deep penetration welding of copper. A recording frequency of 20,000 Hz and a pixel size of 4 μm during the high-speed recordings enabled a high temporal and spatial resolution to evaluate the keyhole geometry. An evaluation of the keyhole in time steps of 0.05 ms was reached, for the first time, and carried out for different geometrical parameters, i.e. width at the surface, depth, maximum bulging and position of the maximum bulging. The unique capabilities revealed that the keyhole behavior was highly dynamic which is why the measured values were further investigated by FFT analysis to determine the occurring frequencies. A novel methodology is developed in this study to show that a mutual influence of radial and axial frequencies was conceivable and individual dominant frequencies can be determined by sine fits. Regarding the keyhole geometry, a comparison of the maximum enlargement of the bulge and its position in depth within the keyhole enabled an evaluation of the shape deviation. It was noticeable that the maximum bulging at the keyhole bottom was most pronounced whereby shape deviations occurred at different positions simultaneously. The results proved the excellent possibilities for gaining an in-depth insight into the complex phenomena of laser beam welding processes of copper based on high-speed and high-resolution synchrotron X-ray imaging. However, the challenges of strong fluctuations and instabilities with conventional intensity distributions come to light clearly.

Based on the presented capabilities and methodologies, further studies will enable a significant scientific and industrial impact since a specific description of the keyhole and its effect on resulting seam imperfections depending different process quantities can be given. Thereby, the intensity distribution is of particular importance and can be manipulated specifically for industrial applications based on the use of concentric intensity distributions, e.g. adjustable ring mode (ARM) technology from COHERENT. The utilization of such intensity distributions allows a significant increase in weld seam quality (see Fig. 8). However, the description of underlying phenomena leading to the stabilization of the process are still unknown which is why further publications regarding concentric intensity distributions utilizing high-speed synchrotron X-ray imaging will follow.

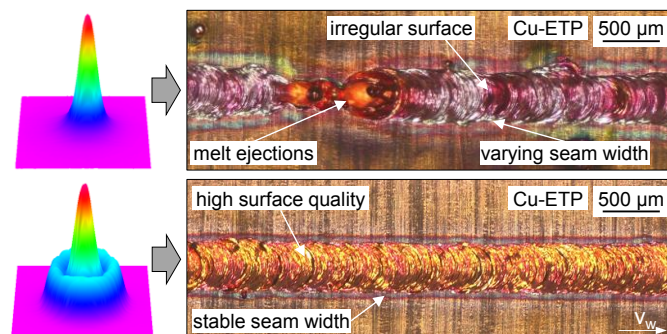


Fig. 8. Weld seam surface for conventional and concentric intensity distributions manufactured by adjustable ring mode (ARM) technology from COHERENT

Acknowledgements

TU Ilmenau and COHERENT thank ESRF – The European Synchrotron for approving the proposal (ME-1580) and the staff of beamline ID19 for their substantial support.

The authors thank the Free State of Thuringia for funding the project “Leistungszentrum InSignA” (2021 FGI 0010) that allowed carrying out the experiments at ESRF.

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